Cloud Computing (SE-490) - Complex Engineering Activity (CEA) - Section B

Appendix A - Deployment Evidence (Screenshots)

Three-Tier Expense Tracker on AWS (ap-southeast-2, Sydney)

Abdur Rafey (SE-23093) · Murtazah Hussain (SE-23097)

Accompanies: CEA Technical Report (separate PDF) · Repository: https://github.com/abdurrafey250904-hash/Expense-Tracker

All AWS Console screenshots were taken in region ap-southeast-2 (Sydney). Additional full-detail AWS CLI exports are in the repository under `docs/evidence/`.

A.1 Network / VPC configuration and subnets

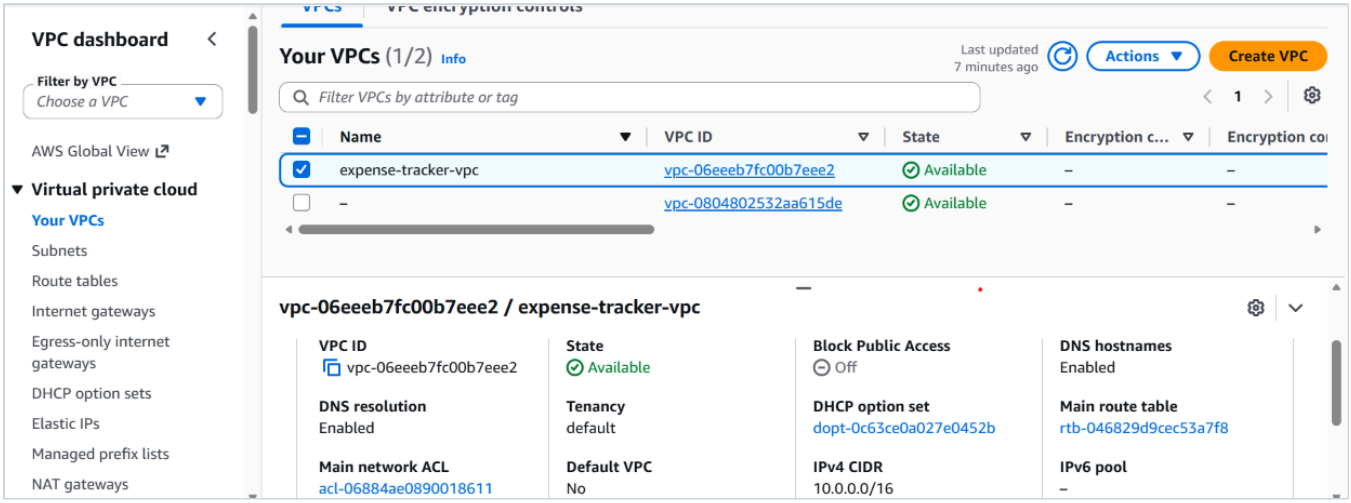


Figure A1 - VPC `expense-tracker-vpc`, IPv4 CIDR 10.0.0.0/16.

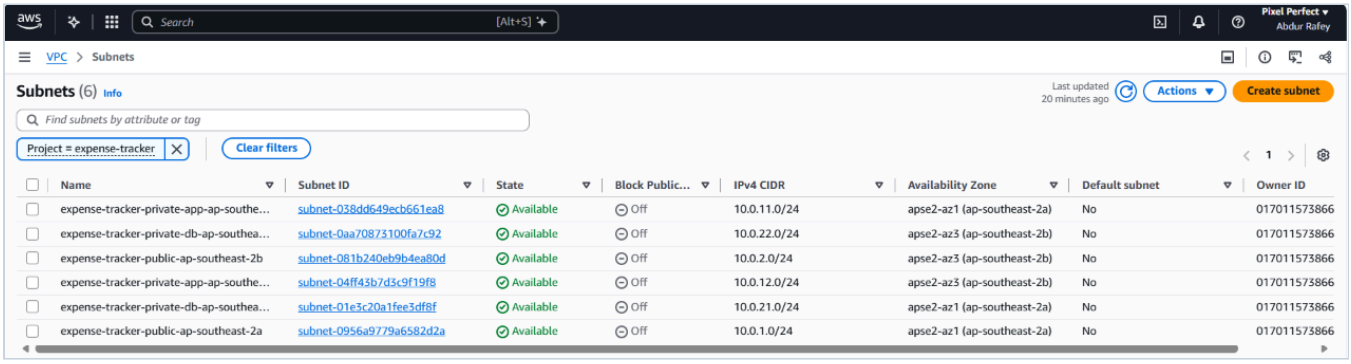


Figure A2 - Six subnets: public, private-app and private-DB in two Availability Zones.

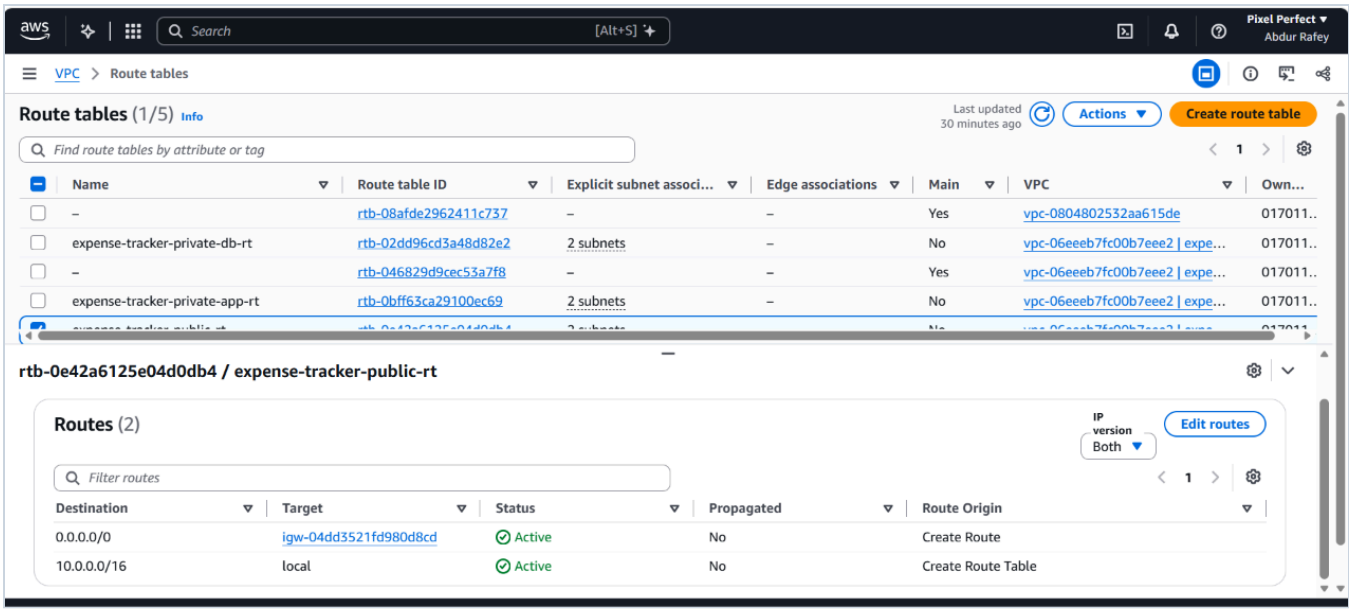


Figure A3 - Public route table: 0.0.0.0/0 → Internet Gateway.

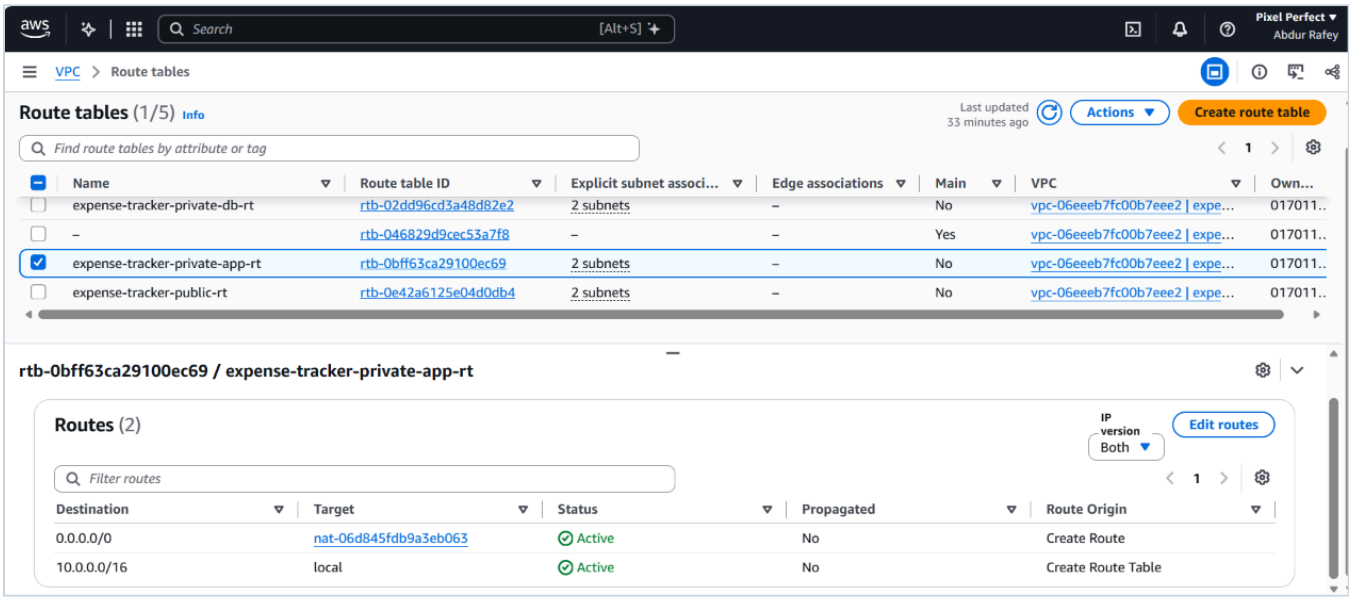


Figure A4 - Private-app route table: 0.0.0.0/0 → NAT Gateway (outbound only).

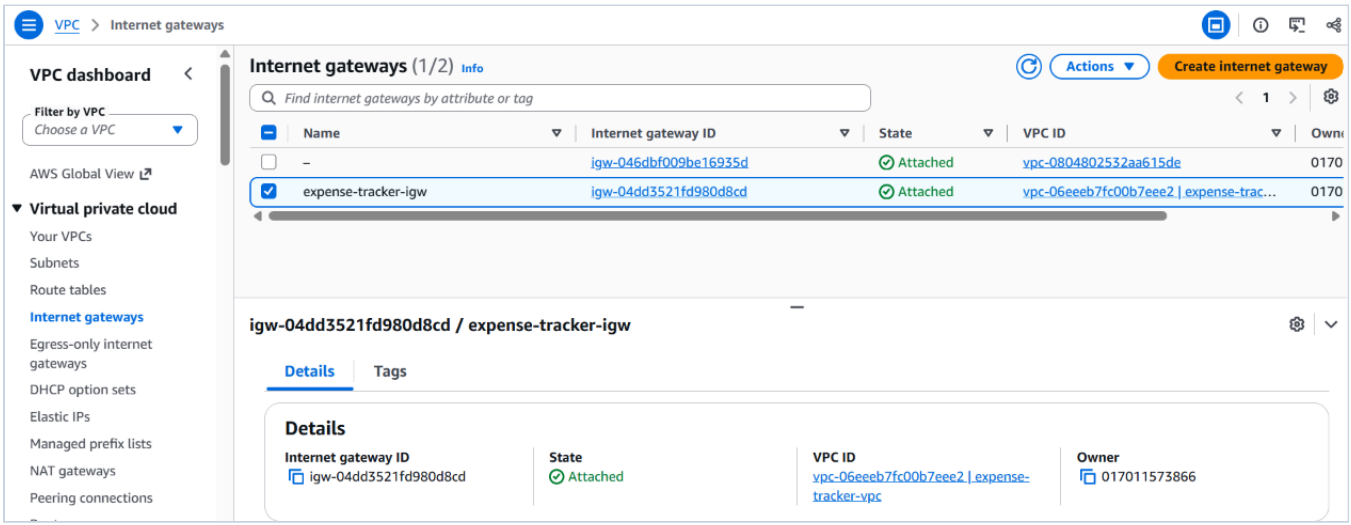


Figure A5 - Internet Gateway attached to the VPC.

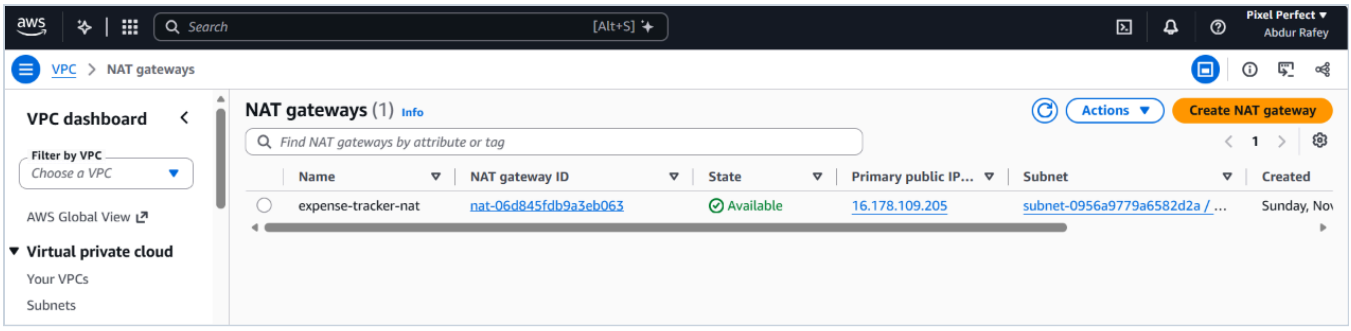


Figure A6 - NAT Gateway (Available) with its public Elastic IP.

A.2 Compute instances / container hosts

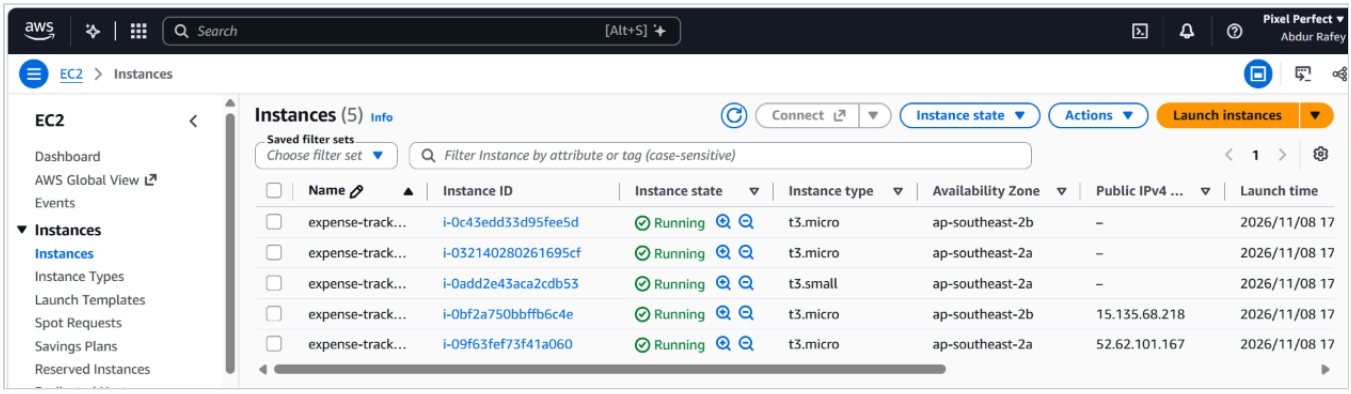


Figure A7 - Five running instances: 2 web (public IPs), 2 app and 1 MongoDB host (no public IPs).

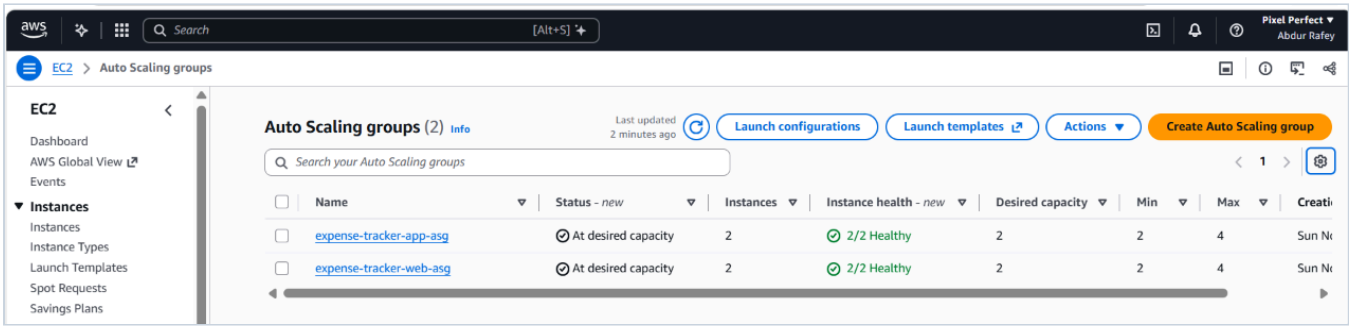


Figure A8 - Web and app Auto Scaling Groups: 2/2 healthy, min 2, max 4.

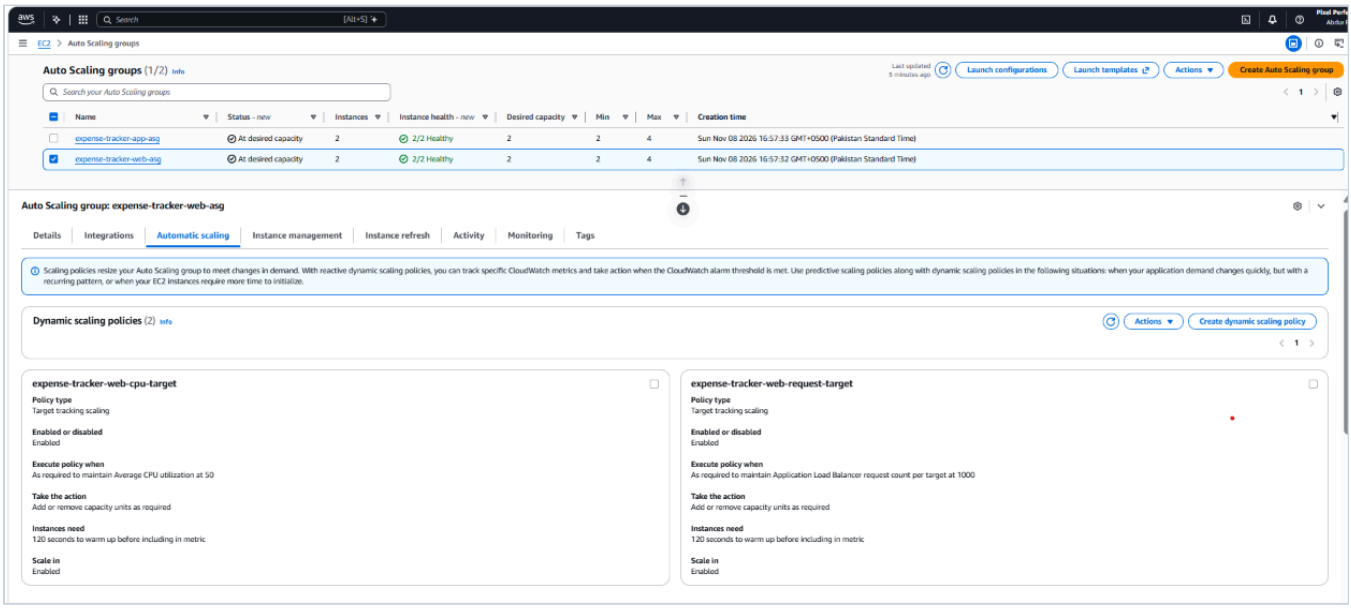


Figure A9 - Target-tracking scaling policies: CPU 50 % and 1,000 requests per target.

A.3 Security groups / firewall rules

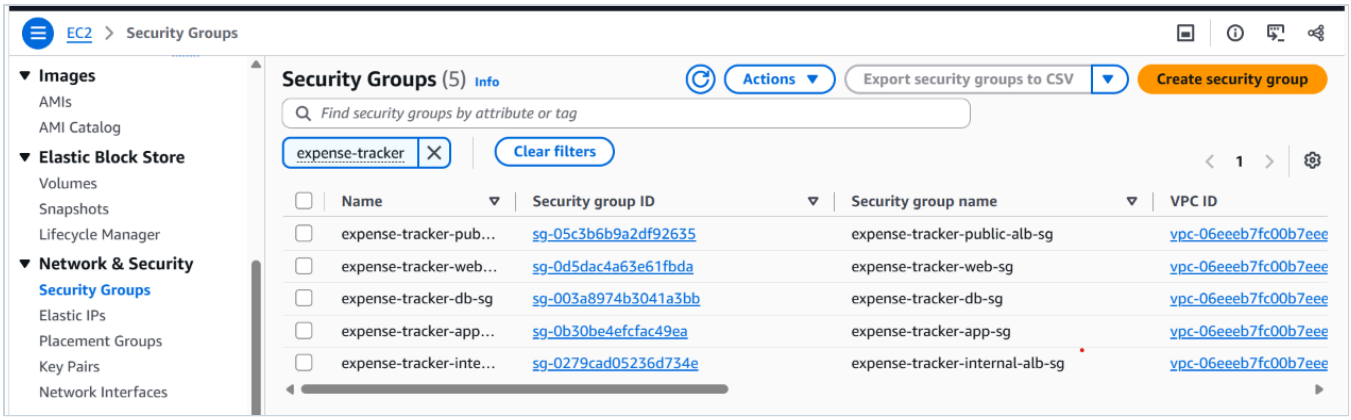


Figure A10 - One security group per tier.

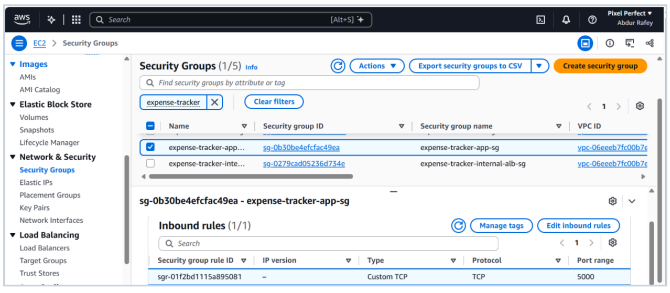


Figure A11 - App SG: inbound TCP 5000 only (source: internal-ALB SG).

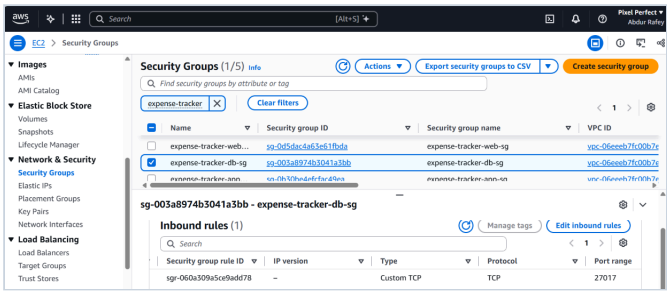


Figure A12 - DB SG: inbound TCP 27017 only (source: app SG).

A.4 Load balancer configuration

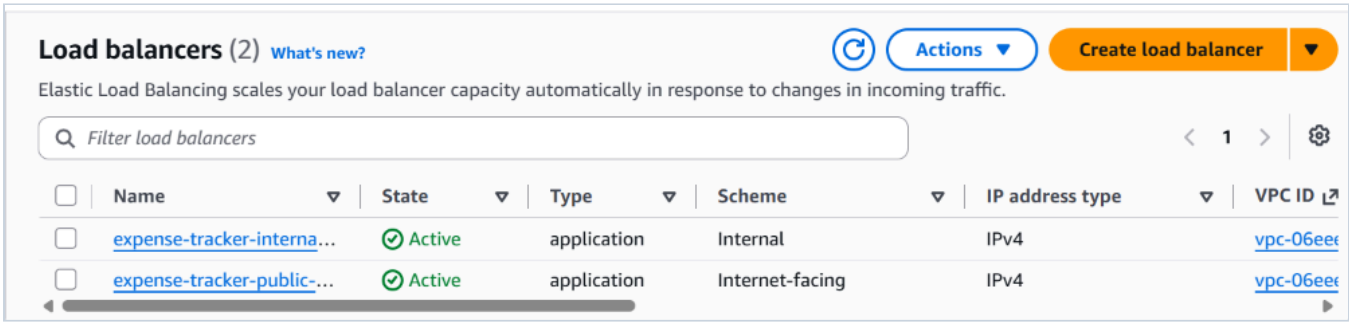


Figure A13 - Internet-facing public ALB and internal ALB, both Active.

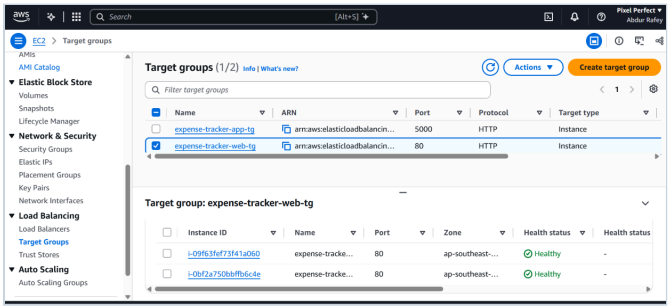


Figure A14 - Web target group: 2 healthy targets on port 80.

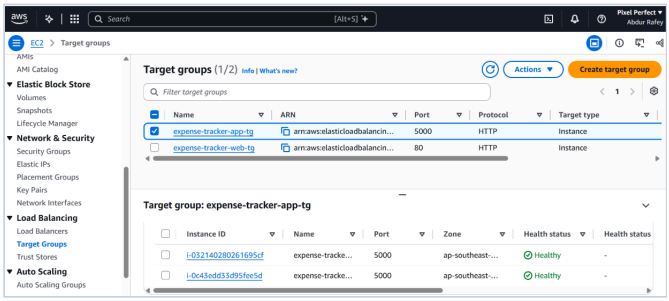


Figure A15 - App target group: 2 healthy targets on port 5000.

A.5 Database instance

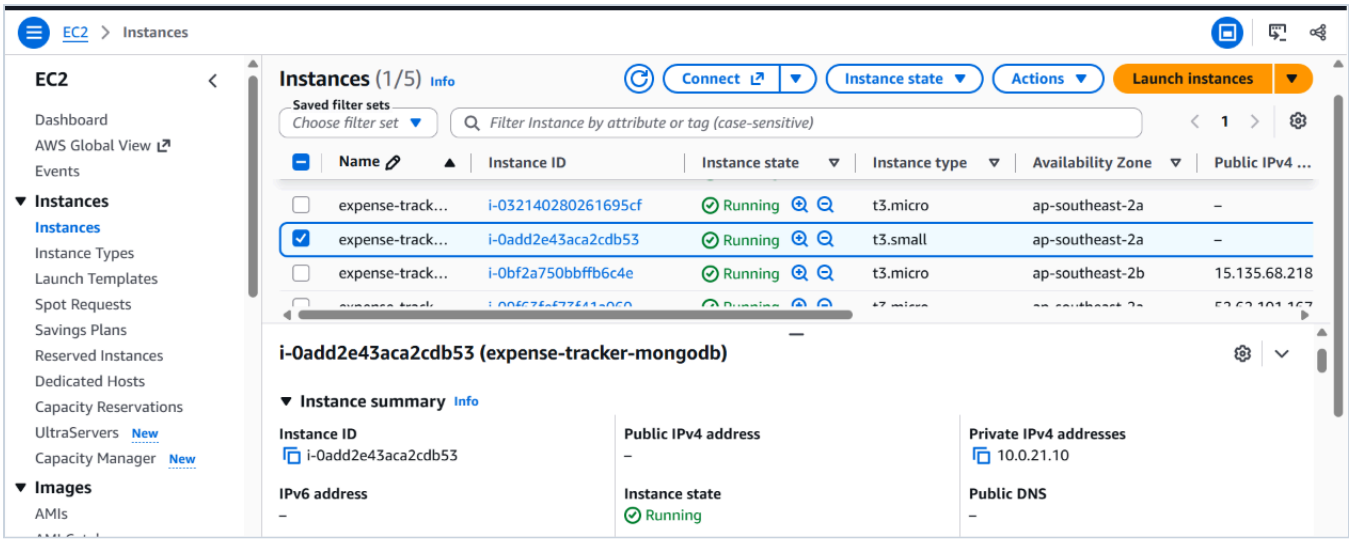


Figure A16 - MongoDB host `expense-tracker-mongodb` : private IP 10.0.21.10, no public IPv4.

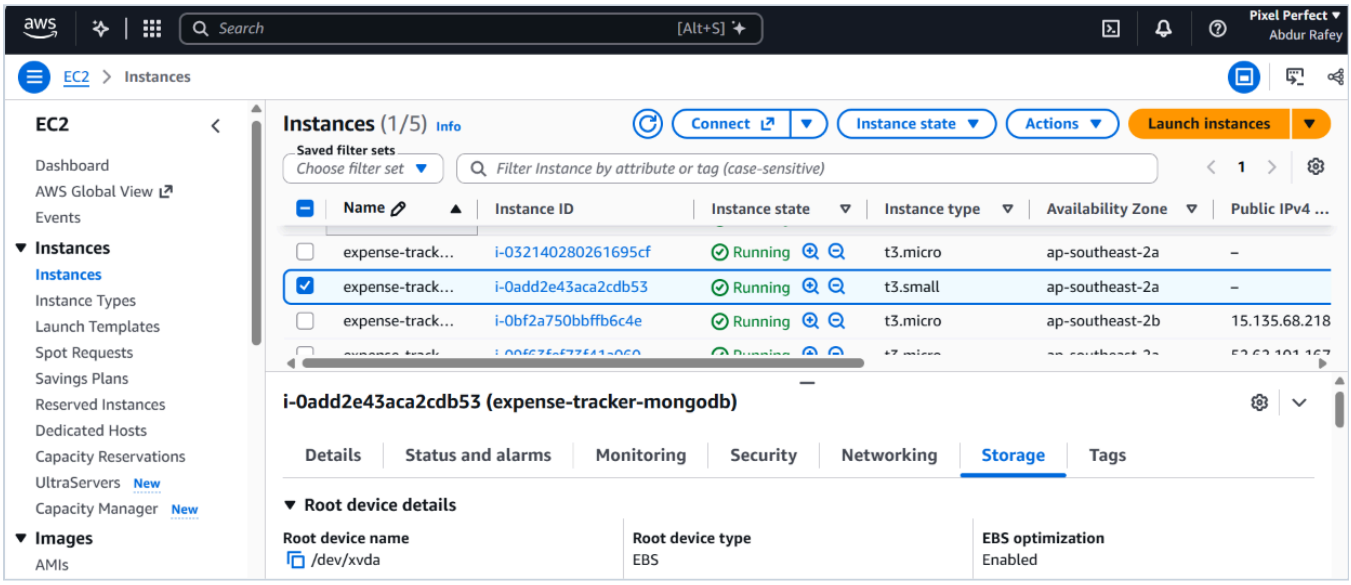


Figure A17 - Database host storage (EBS); the separate 20 GiB encrypted data volume is listed in `docs/evidence/05-database.txt` .

A.6 Docker images and container registry

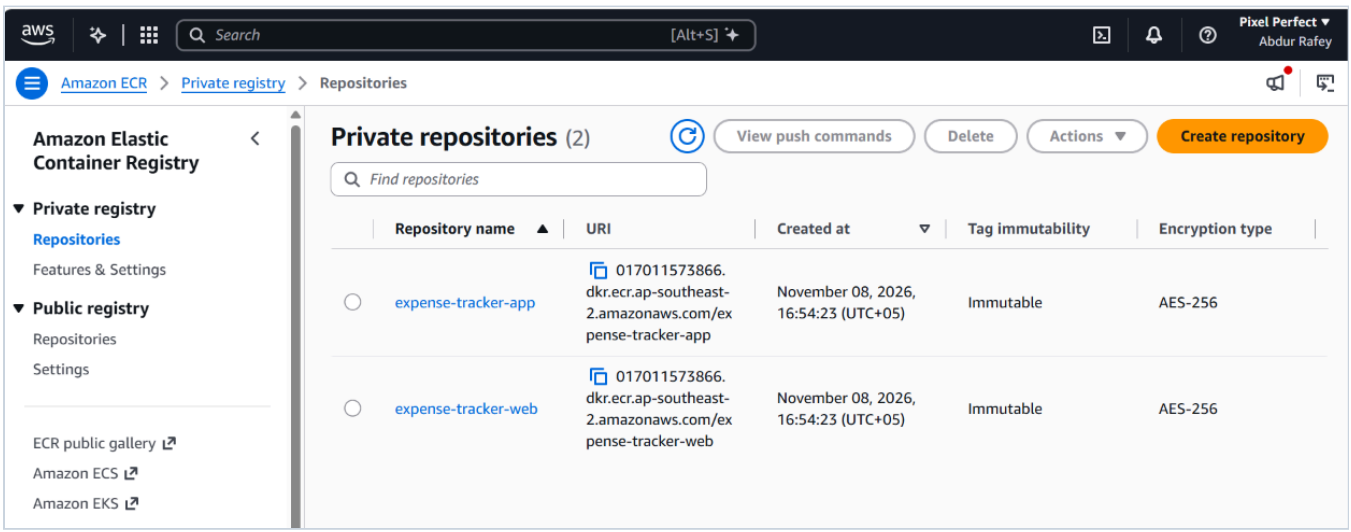


Figure A18 - Amazon ECR private repositories (immutable tags, AES-256).

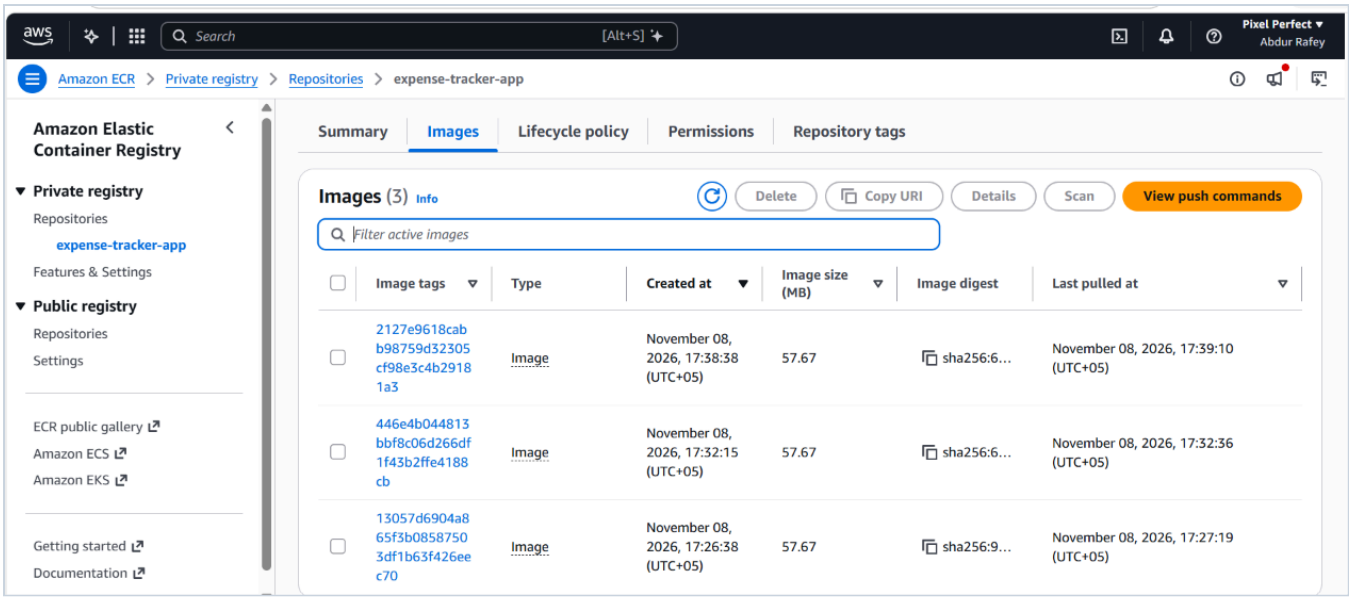


Figure A19 - Images pushed by the pipeline, tagged with git commit SHAs.

A.7 CI/CD pipeline run

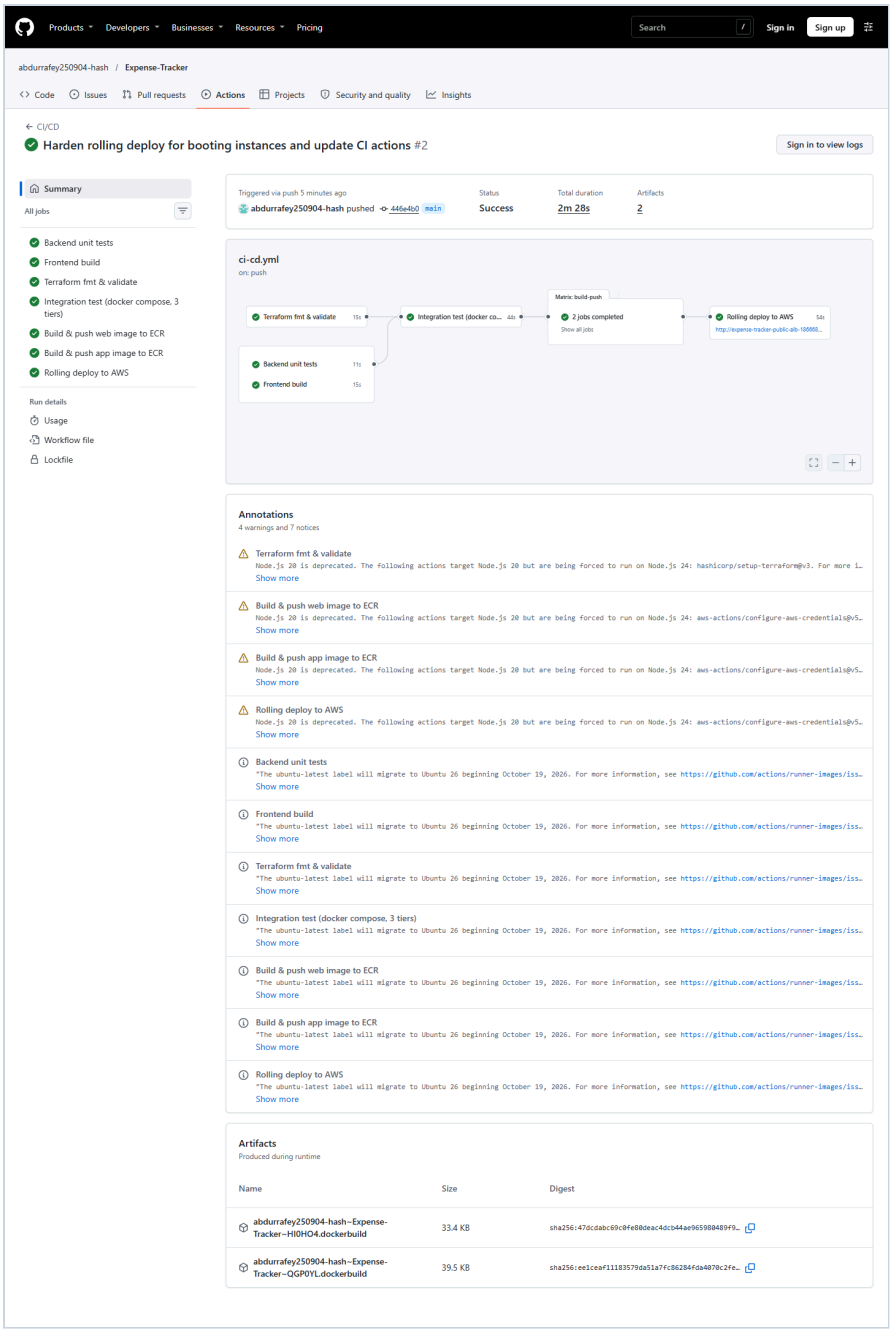


Figure A20 - Successful GitHub Actions run: unit tests, frontend build, Terraform validate → integration test → build & push (matrix) → rolling deploy to AWS. Build-and-push logs: `docs/evidence/06c-docker-build-log-*.txt`.

A.8 Running application accessed via a browser

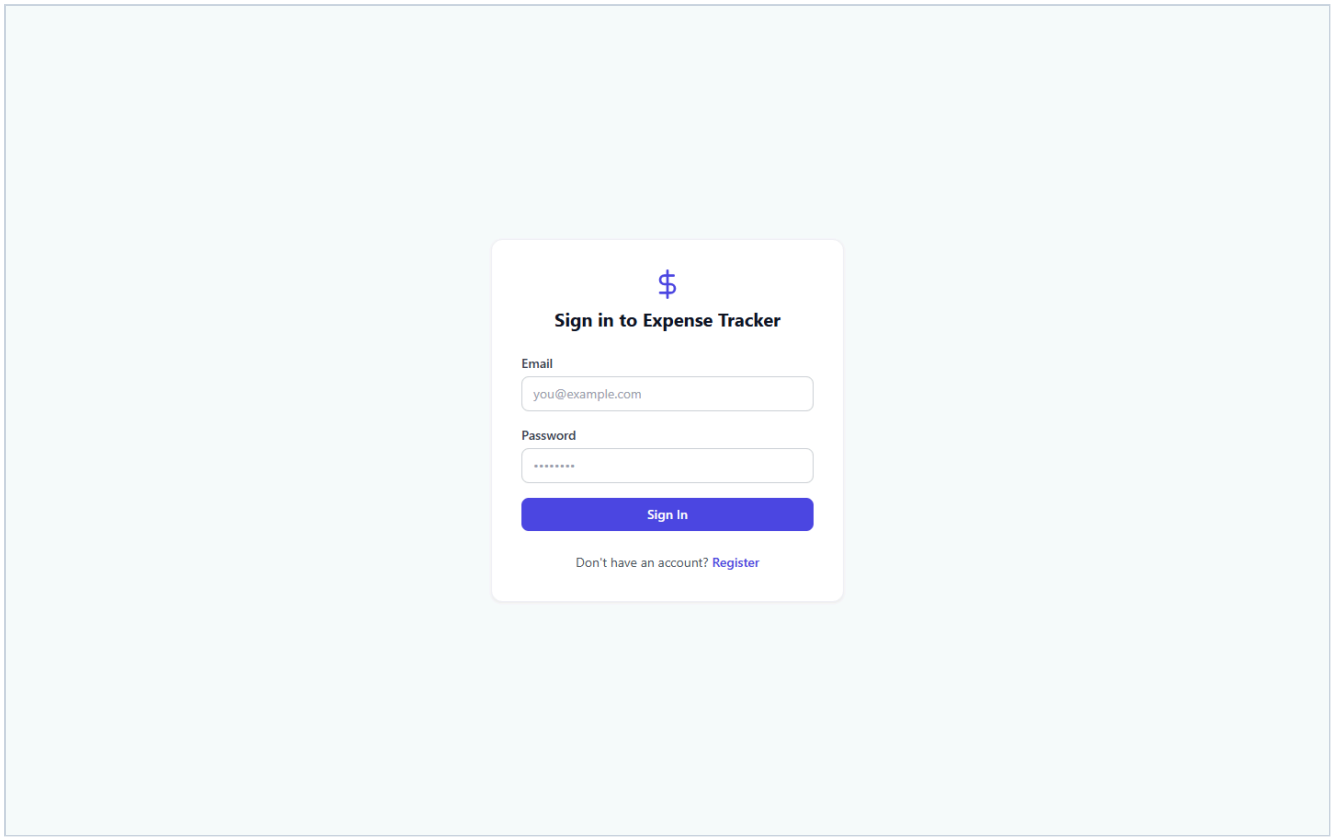


Figure A21 - Login page served through the public load balancer.

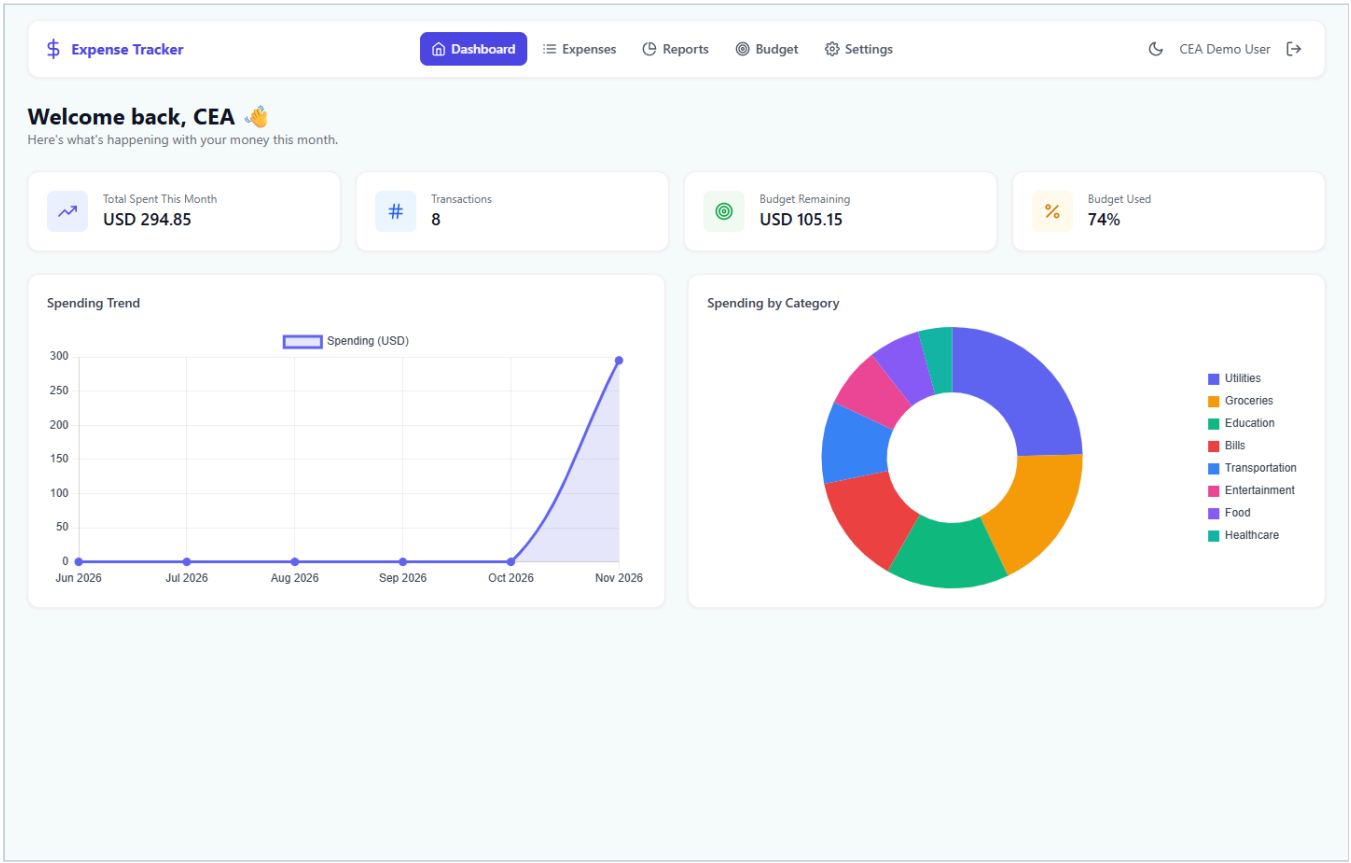


Figure A22 - Dashboard with live data from MongoDB (totals, budget usage, trend and category charts).

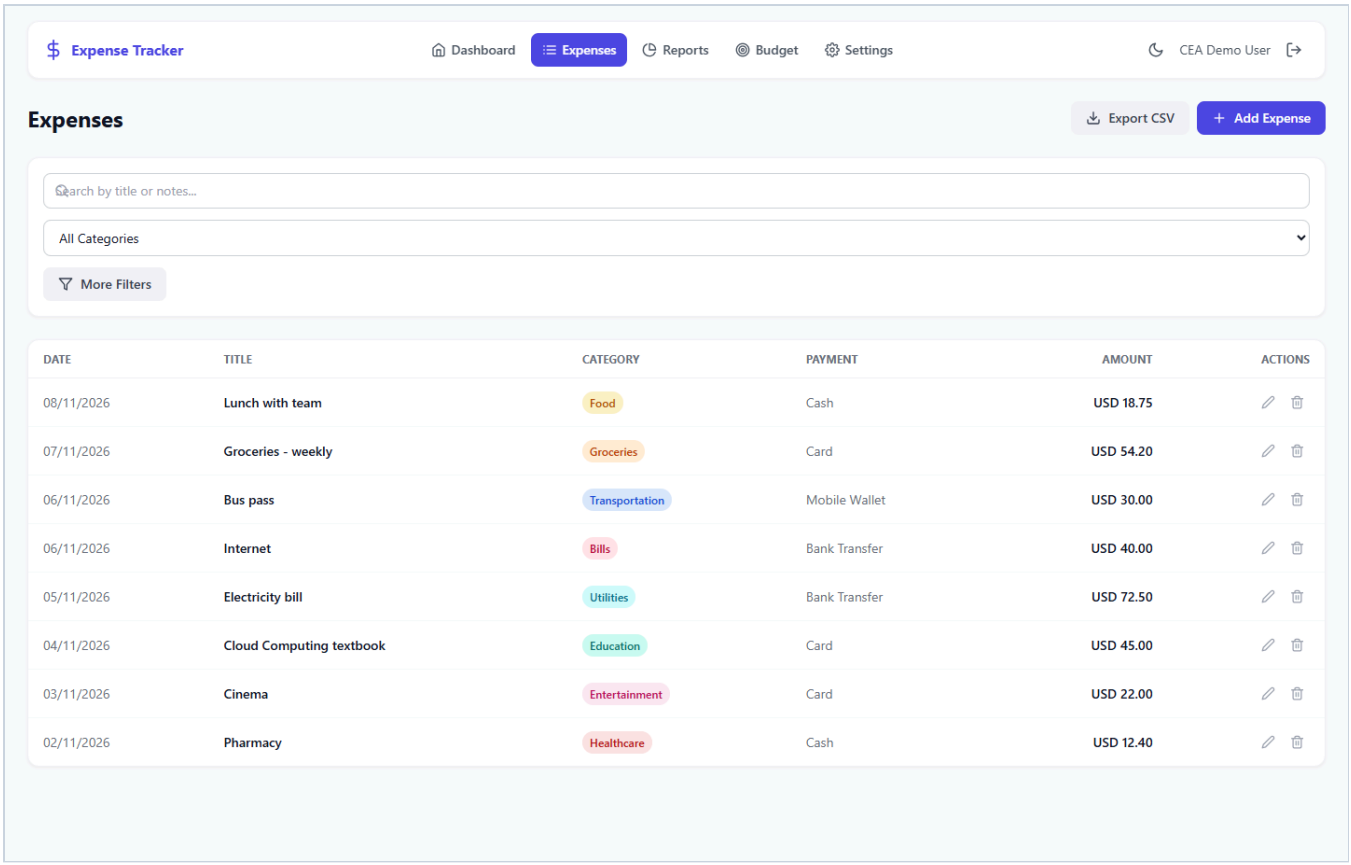


Figure A23 - Expenses list with filtering and pagination.

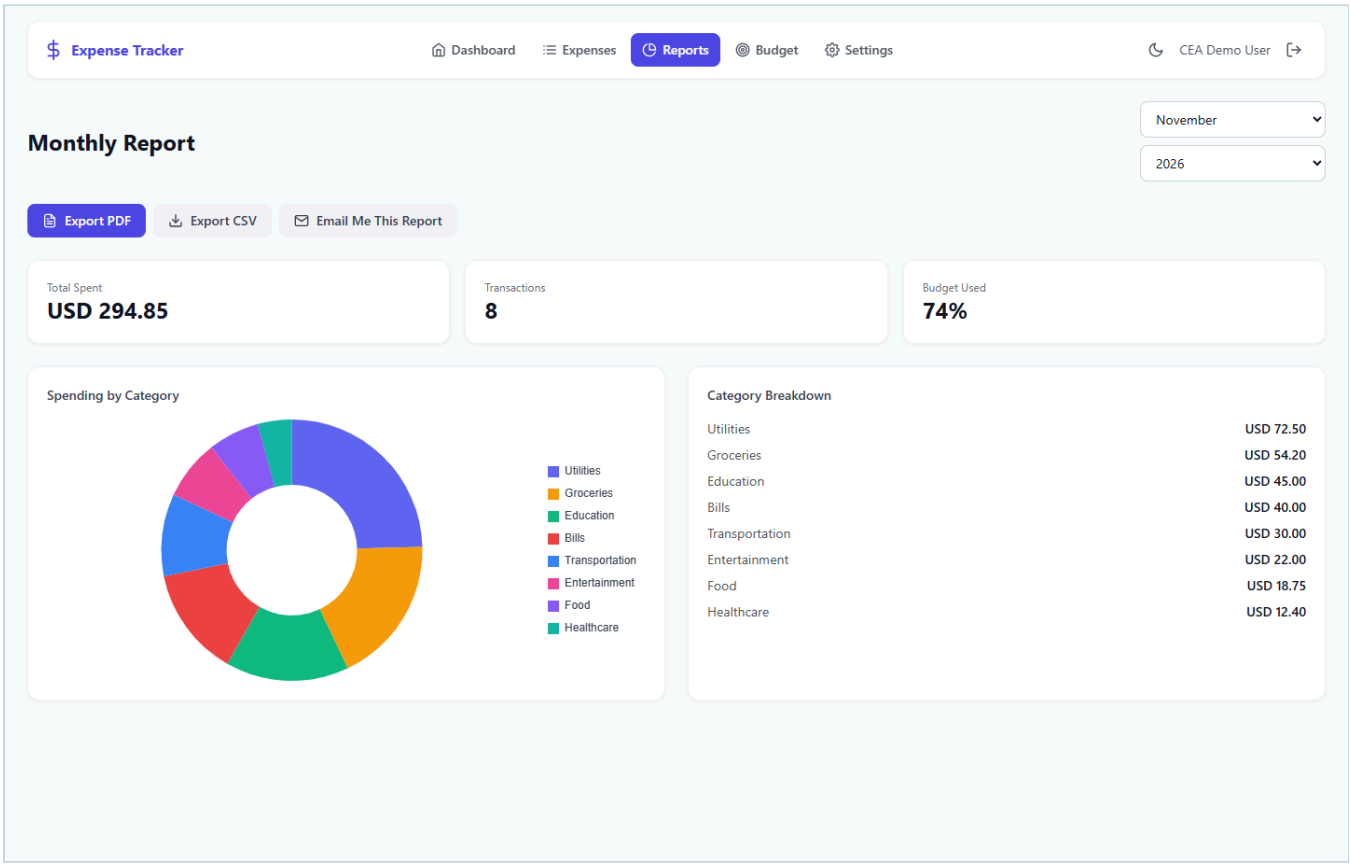


Figure A24 - Monthly report page.

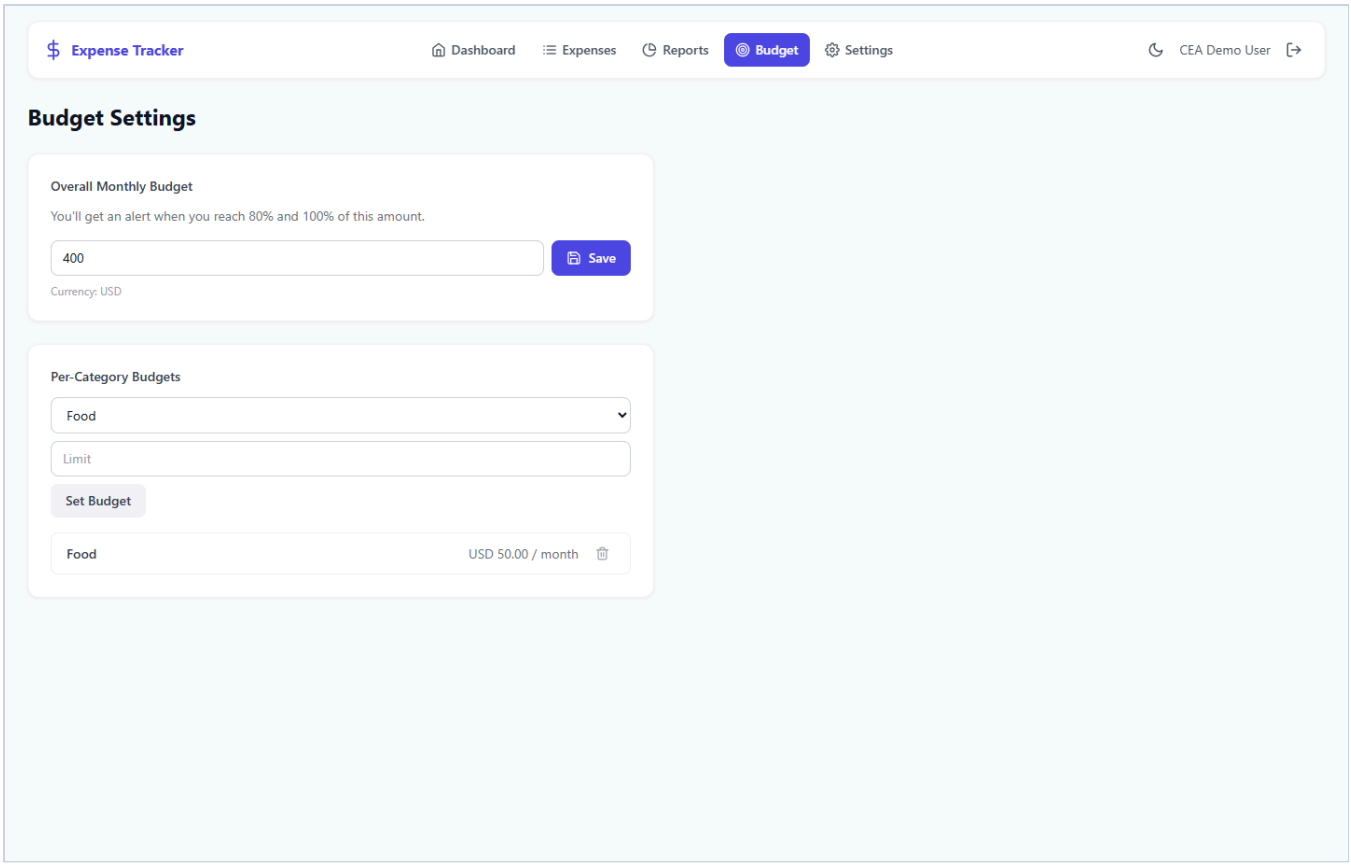


Figure A25 - Budget page with category budget and alerts.

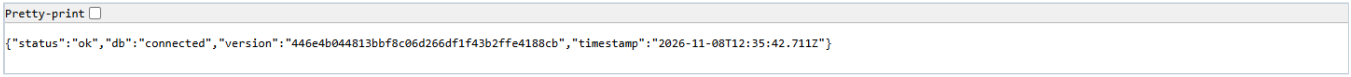


Figure A26 - `/api/health` through all tiers: deployed version and `db: connected`.